1 Study Information

1.1 Title & Authors

Title Climate Policy: Emotions, Attitudes and Trade-Offs

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1.2 Description

This module surveys respondents' attitudes and emotions towards climate change and their support for various mitigation policies (i.e., push- and pull-measures, regulations). Further, it investigates whether citizens trade off democratic principles for their climate policy goals, and how they perceive and solve the trade-off between economic goals and climate action.

2 Theoretical Background

2.1 General Motivation

Climate change is one of this century's greatest challenges and according to 81 % of Europeans even the single biggest one (EIB, 2021). Indeed, 43% of Germans name climate change amongst the top three challenges facing the world already today (DPI, 2022). Problematically, data from the Federal Environmental Agency shows only 26% of Germans think the government and 21% think their fellow citizens are doing enough to prevent climate change (Gellrich, 2020). On the flip side, 18 % of Germans do not even think that climate change is mainly caused by human activities (Peritia, 2022) and support for specific climate change measures can vary wildly. For example, we see generally high support for subsidies for renewable energies, but much less agreement on the taxation of fossil fuels (ESS, 2016)
Taken together, these data points convincingly show the current salience of climate issues, but the story becomes much less clear when looking at the crucial relationship of attitudes, beliefs, emotions and policy support – especially, if climate protection competes with other political goals. With its panel structure, the SOEP is exceptionally positioned to study citizens' positions on climate change, which is why we propose to add our module *Climate Policy: Emotions, Attitudes and Trade-Offs* to the 2023 SOEP Innovation Sample. It combines well-established measures that are not usually fielded in the SOEP which are valuable to a large host of researchers (i.e., general climate belief and attitude items) with novel items investigating economic and democratic trade-offs within climate politics.

2.2 Research Questions

Our proposed module allows us (and others) to study important questions such as which types of climate policies (e.g., push, pull, regulation) receive public support and what beliefs and attitudes drive this support. Beyond that, we are personally, especially interested in the trade-offs between combating climate change and democratic principles and the economy, respectively. We argue that these two trade-offs are important to study for different reasons: Regarding the economic trade-off, previous literature has

identified economic concerns as one of the major deterrents of climate policy support (e.g., Kellstedt et al., 2019; Conrad, 2019; Armingeon and Bürgisser, 2021). Thus, understanding how Germans perceive and possibly resolve this trade-off could greatly inform policy-makers on how to design new policies to attract majority support. While it is widely assumed that citizens prioritize the economy, and in particular their personal economic situation, over climate action, this has rarely been studied at the individual level or with a direct investigation of such a trade-off (e.g., Mildenberger and Leiserowitz, 2017). Investigating this trade-off with the unique data structure of the SOEP allows us to not only take the current economic situations of respondents into account but also look at their economic past.

Regarding the democratic trade-off, Ross Mittiga (2021) recently argued that if climate change is not combated decisively it might pose a danger to democracy in the future by pitting questions of survival against the upholding of democratic norms. We agree with this general assessment, as it is known from other literature (e.g., on terrorism or migration) that citizens might trade off civil liberties to protect themselves from a perceived threat (e.g., Dunwoody and McFarland, 2018; Shepherd et al., 2018). In contrast to Mittiga, we believe these trade-off considerations are already present in the current society (even if it is in subtle ways). Supporting this claim, Graham and Svolik (2020) showed for the U.S. that many voters value partisanship and policy goals (including environmental ones) over democratic principles in their electoral choices. However, there is barely any study evaluating direct effects of climate threat on commitment to democratic norms with regards to climate politics (but see Amat et al., 2020, for evidence from Spain). Understanding how climate change can already today enlighten undemocratic sentiment through associated emotions is crucial to safeguard social cohesion and democratic institutions. We thus want to provide first comprehensive evidence about this mechanism in Germany.

2.3 Theoretical Background for singular items (see also list in 5.2)

- **General Climate Attitudes & Beliefs** This section most importantly provides valuable control variables for all our analyses. Due to time considerations, we limit it to three items that are fundamental to climate change attitude formation: existence, natural vs. man-made & perceived knowledge about climate change (see Drews and van den Bergh, 2016; Valkengoed and Steg, 2019, for meta-analyses about predictors of climate policy support and adaptive behavior).[1]

- **Climate Politics & Policy Preferences** In order to measure citizens' abstract support for (or rejection of) climate change mitigation measures we use a slightly modified item from the *GLES Pre-Election* questionnaire (GLES, 2022). Since abstract support does not necessary translate into support for specific policies, we then ask about eight specific policies following both a logic of scientific and relevancy. Scientifically speaking, we first aim to cover the three most relevant *types* of policies generally discussed in the literature (e.g., Huber and Wicki, 2021; Wicki and Fesenfeld, 2019): market-based pull-measures that provide incentives, push-measures that provide deterrents, and non market-based regulations that introduce strict rules.[2] To this we add a fourth category, preliminary labeled as "civil rights", which includes policies that actively undercut civil liberties in order to further climate protection. This category enables us to study which citizens are willing to trade off civil liberties for climate protection policies.[3] Since previous research showed that perceived effectiveness (e.g., Huber et al., 2020; Singh et al., 2017) and subjectively expected financial benefits or costs from a policy (e.g., Zahran et al., 2006) affect policy support, we ask

[1] Other possibly useful items include left-right self-placement, social and political trust (e.g., Kulin et al., 2021; Cologna and Siegrist, 2020) or perceived scientific (e.g. Ding et al., 2011) or social (e.g., Goldberg et al., 2019) consensus.

[2] We are aware that the planned "Umweltmodul" in the 2023 SOEP plans to use similar categories to survey general agreement with these policies types. However, we believe there is still more to be learned by taking up explicit contemporary policy proposals that fit into this typology.

[3] Note, that we include two policies per category to limit the risk of idiosyncratic policy effects.

respondents to evaluate all policies from these three points of view.[4] Especially the latter relates directly to economic trade-offs and should differ from "objective" impact measures scholars could deduce from respondents' socio-economic circumstances (see Jagers and Hammar, 2009).

- **Economic Trade-Offs** One of the major interests of our research is how respondents perceive and (if necessary) resolve the trade-off between economic and climate action goals. Previous research finds that economic concerns are one of the most important policy goals that need reconciliation with climate goals (Mildenberger and Leiserowitz, 2017). The longitudinal individual- and household-data of the SOEP provides an invaluable data advantage over most existing research as it allows to take personal financial situations and developments into account more thoroughly. In addition, we add a more abstract measure asking about citizens' perception whether combating climate change hurts the German economy and labour market in both the long- and short-term. Most importantly, we then present respondents with a hypothetical scenario in which they have to choose which goal (economic of ecologic) they put over the other. Existing research does not ask trade-off questions, but only studies how public opinion on climate action and support for climate policies changes in financially difficult times, and/or is not examining the data on an individual level (e.g., Brulle et al., 2012; Kachi et al., 2015; Kahn and Kotchen, 2011), or when the cost of the policy is emphasized (e.g., Aklin and Urpelainen, 2013; Ansolabehere and Konisky, 2014). Furthermore, by surveying respondents' attitudes towards the long- and short-term effects of climate change mitigation measures, we are able to disentangle if trade-offs and policy support are dependent on the actual, personal financial situation or on perceptions about the future economic development. Therefore, these items improve both the measurement and the level of analysis for studying the trade-off between the economy and climate action significantly.

- **Democratic Trade-Offs** As touched upon in 2.2 these items attempt to measure whether citizens are willing to trade off democratic values or civil liberties for their preferred climate politics. To our knowledge, there is no direct example of existing literature to model these items after. We address this by measuring the trade-off in multiple ways. Firstly, through the two *civil rights* policies probing whether citizens are willing to support policies that severely restrict either freedom of speech of property rights. Secondly, we include a battery of binary-items surveying approval for more or less undemocratic statements, this time targeting both sides of the climate cleavage. We model these after current debates.[5] Importantly, all statements are, in our opinion, in some way detrimental to a functioning democracy (i.e., disregard of courts, technocracy instead of representative democracy, limiting freedom of opinion or protests[6]). Compared to the economic policy-goal conflict, the democratic trade-off is much more value-laden. Support for democracy might be susceptible to misreporting errors such as social desirability (e.g., Kiewiet de Jonge, 2016), potentially leading to respondents underreporting their willingness to transgress democratic norms. Therefore, we lastly also employ a two-step list experiment (Graeme and Imai, 2012) to identify and correct for misreporting. The sensitive items used in this list experiment are also included in the previous battery of binary approval-items allowing us to not only estimate how many people *actually* agree with this statement, but also which sections of society are willing to state this publicly. This sheds light onto the validity of the other, direct survey approaches and helps to avoid underestimation of the threat posited by undemocratic sentiment.

[4] Based on previous literature one might also consider, for example, surveying which policies respondents consider fair (e.g., Hammar and Jagers, 2007).

[5] For example, the statement asking for climate activists to be prosecuted echoes recent comments made by CDU-general secretary Mario Czaja (DPA/ZDF, 2022).

[6] How *undemocratic* these statements really are is hard to state a priori. One way to assess this would be to include an additional measure in a subsequent wave that asks respondents to rate the same statements according to their level of danger to democracy.

- **Psychological Distance & Emotions** Previous literature has linked psychological distance [PD] (e.g., Wang et al., 2019; Maiella et al., 2020; Singh et al., 2017) and emotions (e.g., Wang et al., 2018; Smith and Leiserowitz, 2014) to environmental behavior and policy support. We propose that both concepts also influence the so-far largely unstudied trade-offs at the core of our proposal. The concept of PD can be traced back to Yaacov Trope and Nira Liberman (2008; 2010), and posits that people evaluate situations (and threats) differently based on how close or distant they feel to them. The current conception of PD consists of four highly correlated dimensions: spatial distance (e.g., how far away the threat is), social distance (e.g., how different the target is from oneself), temporal distance (e.g., when this threat will manifest) and hypotheticality (e.g., how certain this threat will manifest) (Singh et al., 2017). We argue, the closer respondents perceive harmful climate change effects psychologically, the more willing they are to trade off economic or democratic values for climate protection. To measure PD, we distilled six items out of existing measures (e.g., Spence et al., 2012; Singh et al., 2017) that should convincingly capture all dimensions.
  Finally, we look at seven distinct emotions with different hypothesized effects on trade-off considerations: *Anxiety*, which is triggered in situations in which individuals perceive low self-control, can reduce political participation (Wagner and Morisi, 2019). Meanwhile, *anger* serves as a motivation to correct a problematic situation (van Doorn, 2018) and has been shown to increase partisan tendencies, and to encourage political participation (Valentino et al., 2011). Therefore, we expect that anxious respondents are less willing and angry respondents are more willing to engage in trade-offs according to their partisan position. The other emotions are mostly selected based on a reading of recent literature that deals with climate change and emotions: *hopelessness* decreases policy support (e.g., Mayer and Smith, 2019) and we expect it to prevent trade-offs; *guilt* exists prominently amongst younger citizens but has unclear effects on policy support (e.g., Chu and Yang, 2019; Smith and Leiserowitz, 2014); and the positive emotions *calmness* and *confidence* can foster environmental collective action (Schneider et al., 2021), but make climate protection less urgent and could thus inhibit trade-off tendencies. Finally, we include the so-far neglected emotion of *annoyance*. To our knowledge, this has not been studied before, but it is an emotion that we believe is tremendously prevalent amongst the public on both sides of the issue. We measure these emotions once with regards to climate change effects and once to climate change politics in particular. This differentiation allows us to also draw insights about people, who do not care a lot about climate change, but react negatively to any proposed policies combating climate change.

3 Design Plan

3.1 Study type

The proposed module contains one experimental element (a list experiment, incl. the respective direct questioning - see 3.4). The rest of the module follows a non-experimental, observational logic.

3.2 Study design

Our module is designed in a cross-sectional logic and in order to answer our primary research questions about trade-off considerations we only need the items to be included in one wave. However, given the importance of climate change, we think that regular inclusion of some of the more "basic" items (i.e., General Climate Attitudes, Beliefs & Policy Preferences) would be a valuable addition to the SOEP for many researchers.

For the list experiment and the related direct questioning (three affected questions), respondents need

to be randomly assigned into two treatment groups.

3.3 Hypotheses

- **Hyp 1: (non-directional)** People are generally willing to trade off democratic principles for their preferred climate politics.
- **Hyp 1a: (directional)** Anxiety, hopelessness, calmness, and confidence about climate change decreases people's willingness to trade off democratic principles.
- **Hyp 1b: (directional)** Anger about climate change effects increases people's willingness to trade off democratic principles, but anger at the government reduces it.
- **Hyp 1c (directional):** Psychological closeness to climate change increases people's willingness to trade off democratic principles.
- **Hyp 2: (directional)** People misreport their willingness to trade off democratic principles due to social norms.
- **Hyp 3: (non-directional)** People are generally willing to trade-off climate change mitigation for economic growth.
- **Hyp 3a: (directional)** Negative expectations about long-term consequences of measures to combat climate change increases people's willingness to trade-off climate change mitigation for economic growth.
- **Hyp 3b: (directional)** Negative expectations about the short-term consequences of climate change mitigation measures only increase the willingness of economically insecure people to trade climate protection for economic growth.
- **Hyp 4: (directional)** The worse people's personal economic situation, the lower the probability that they support climate policies.
- **Hyp 5: (directional)** Perceived effectiveness of climate policies increases the probability that people support climate policies even if they believe that these policies would bring personal financial disadvantages.

3.4 Randomization

For the list experiment and the related direct questioning (three affected questions), every individual respondent needs to be randomly assigned into one of two treatment groups. Assignment should be done completely by random and with equal probability.

In the list experiment, respondents selected in **Group I** will first receive the sensitive item "If necessary, stronger climate control measures should be prevented by undemocratic means" with a set of three control items, those in **Group II** receive "If necessary, stronger climate control measures should be introduced by undemocratic means" with a different set of three control items (see also 9.1 & 9.2). The control items are selected from a GLES attitude battery in a way to avoid floor or ceiling effects (GLES, 2022). In a second step the groups receive the set of three control items that was not included with their sensitive item. Simply put: The two halves serve as control group for each other. In the last question of the whole proposed module (i.e., the direct questioning of democratic trade-offs), all respondents receive four general items plus their own sensitive statement from the list experiment.

4 Sampling Plan

4.1 Sample size & sample size rationale

There is no clustering or multilevel analysis planned. According to the survey report on the 2020 SOEP Innovation Sample (Zweck and Rathje, 2021), 3.785 individual interviews were completed. Since we do not have any exclusion rules, we treat this as our potential (maximum) sample size. To give an estimation about our minimally necessary sample size we conducted power analyses for a logistic regression (Fig.1, corresponding to any binary agreement items as dependent variables) and two for five-step ordered logistic regressions (Fig.2 & 3, corresponding, for example, to policy support as dependent variable). For the latter, we conducted two tests in which we varied the probability of respondents to fall into the singular categories, as this can have a negative influence on power if it is very uneven. Note, that all power analyses presume an alpha-level of 0.05.

Figure 1: Power-Analysis for Logistic Regressions - performed in *G*Power3*

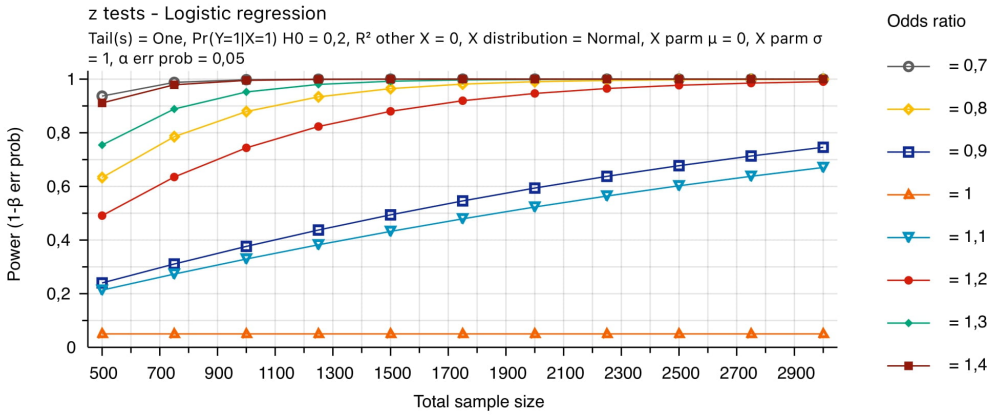


Figure 2: Power Estimation for Ordered Logit using the *Hmisc*-function in R - based on five categories with equal probabilities (p = 0.2 each)

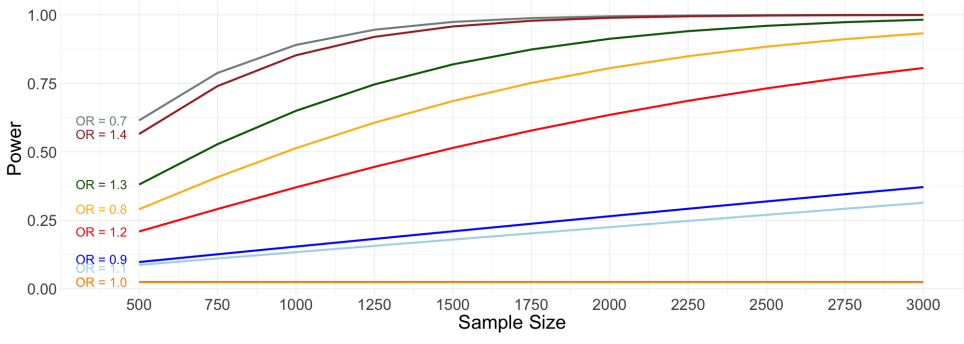


Figure 3: Power Estimation for Ordered Logit using the *Hmisc*-function in R - - based on five categories with different probabilities (p = .05; .5; .2; .15; .1)

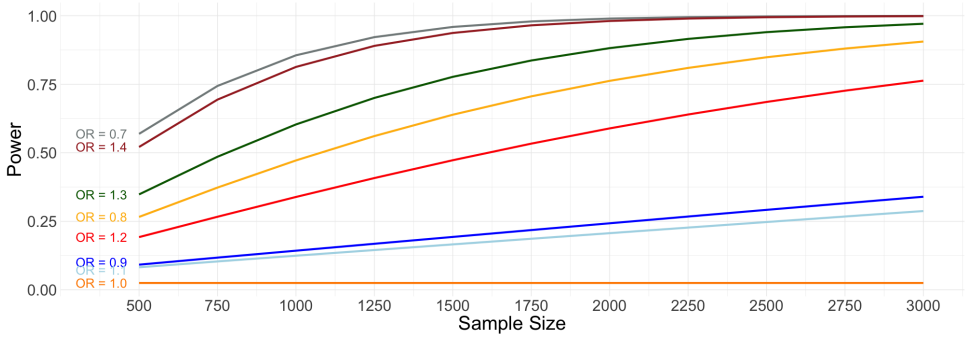


Fig.1 suggests that any effect with odds ratios [OR] above 1.2 or below 0.8 can be identified with a strong power ($\geq$ 0.8) with about 2.000 respondents. However, only if we increase the number of respondents to 3.000, are we able to identify effects with OR of 0.9 or 1.1 with a moderate power (ca. 0.6). Generally speaking, Fig.3 shows that we have slightly less power assuming a strongly uneven allocation into categories, but even in this case 1.750 respondents are enough to obtain a power above 0.5 for any OR above 1.2 or below 0.8. To get a comparably good power ($\geq$ 0.75) for the same effect strengths, we would need about 3.000 respondents. Meanwhile, the sample does not really provide enough power ($\leq$ 0.25) to find any effects with ORs between 0.9 or 1.1.

Conducting a power analysis for the list experiment is less straightforward, as power heavily depends on the probability with which people affirm both, the sensitive and the control items (Graeme and Imai, 2012). Following the advice of these authors, we selected our control items to maximize power and based on their simulation study, we argue that a minimum of 1.000 respondents per randomized group (2.000 in total) is necessary to provide sufficient power to the experiment (Graeme and Imai, 2012, p.71).

Taken together, theses analyses suggest that a sample size of 2.000 individuals would be a minimum requirement for this module, with 3.000 respondents being our ideal target. Since understanding especially the economic trade-off questions could be greatly enhanced if we know respondent's past financial and job developments, it would be beneficiary to administer our module to respondents from all, but especially the longer participating sub-samples within the Innovation Sample (i.e., Sub-Samples I1, IE, I2, I3) (Zweck and Rathje, 2021).

5 Variables

5.1 Manipulated variables

Since the only experimental variation is a simple randomization into two groups for the list experiment (see *Measured Variables* below) the only "manipulated" variable that needs to be captured is which of the two groups respondents were assigned to. This randomization will affect the first two questions (the list experiment) and the last question (the direct questioning of the sensitive item) in the module.

5.2 Measured variables

Our module is grouped in distinct sections that are ordered in the following way:

1. Democratic Trade-Offs: List Experiment

- **List Experiment - Sensitive I/II** as well as the corresponding **List Experiment - Control I/II** items are targeted at eliciting respondents' true (*hidden*), abstract support of using undemocratic measures to further their own climate policy agenda. Respondents have to indicate how many (not which) out of 4 statements (3 in the control questions) they support.

2. General Climate Attitudes & Beliefs

   - **Existence** captures on an ordinal scale whether respondents belief climate change exists or not.
   - **Cause** captures on a scale from 1 - 11 what respondents perceive as the cause of climate change (while including an option to state that there is no climate change). The scale ranges from climate change being only caused by natural processes (1), over equal importance of natural processes and human activities (6) to being fully caused by humans (11).
   - **Information** captures on a ordinal scale how well informed about climate change citizens perceive themselves to be.

3. Climate Politics & Policy Preferences

   - **Climate Politics - Attitudes** captures on a scale from 1-11 whether respondents generally want the government to take more (1) or less (11) action to combat climate change.
   - **Policy ...** captures support for eight distinct policies proposed to combat climate change on an ordinal scale (for the logic behind their selection see *Theoretical Background*).
   - **Benefits** captures binary from which of the eight policies respondents expect to personally benefit financially.
   - **Costs** captures binary from which of the eight policies respondents expect personal financial costs.
   - **Effectiveness** captures binary which of the eight policies respondents deem effective to fight climate change.

4. Economic Trade-Offs

   - **Economy - Short-Term** captures on an ordinal scale whether respondents think measures combating climate change will be hurtful, beneficiary or without impact for economic growth and the labour market in Germany in the short run.
   - **Economy - Long-Term** captures the same, but for long term effects.
   - **Economy - Trade-Off** forces respondents to (binary) choose whether they would prioritize protecting economic growth or combating climate change if those two policy goals were to contradict each other.

5. Psychological Distance & Emotions

   - **PD - Personal** captures on an ordinal scale how strongly respondents expect climate change to affect themselves personally.
   - **PD - Local** captures on an ordinal scale how strongly respondents expect climate change to affect their local area.
   - **PD - Country** captures on an ordinal scale how strongly respondents expect climate change to affect Germany.

- **PD - World** captures on an ordinal scale how strongly respondents expect climate change to affect the global south.
- **PD - Timing Germany** captures respondents' belief how soon climate change effects will be noticeable in Germany. The item is measured in an open answer format requiring respondents to state a timespan in years, while simultaneously providing them the opportunity to state that (a) effects are already noticeable today, or (b) that they never will be.
- **PD - Timing World** is worded equivalently to the previous item, but asks about effects worldwide instead of being limited to Germany.
- **Emotions - Climate Change** captures how strong or weak respondents feel a selection of seven emotions (for selection logic see *Theoretical Background*) when they think about climate change and its effects.
- **Emotions - Government** captures how strong or weak respondents feel the same selection of seven emotions when they think about the current politics regarding climate change.

6. Democratic Trade-Offs: Direct

   - **Undemocratic - Direct** captures binary agreement or disagreement with a series of five statements that all (to varying degree) imply undercutting democratic values or processes for climate policy goals. These items also include the respective sensitive item evaluated in the list experiment.

It is important to note, that we do not think our proposed items necessarily have to be included in one cohesive block within the survey. Instead, the *General Climate Attitudes & Beliefs*, *Climate Politics & Policy Preferences* and *Economic Trade-Offs* sections would fit perfectly to the already planned "Umweltmodul" section. Meanwhile, the *Psychological Distance & Emotions* and *Democratic Trade-Offs: Direct* section could be placed somewhere else. Importantly, the *Democratic Trade-Offs: List Experiment* needs to be positioned before and sufficiently far away from the direct questioning of democratic trade-offs in order to avoid priming effects. As it includes control items relating to many different topics, it should not be to difficult to fit it into the survey flow.

6 Analysis Plan

6.1 Statistical models

Since we approach the democratic trade-off hypotheses in three distinct ways, we also employ three different statistical models to test this argument: Hypotheses 1, 1a, 1b, and 1c will be first operationalized with the binary agreement item battery asking about undemocratic practices to achieve climate policy goals. Hence, we conduct a series of logistic regressions for each item using agreement as dependent variable, and the respective variables mentioned in the hypotheses (i.e., emotions, psychological distance) as predictors. Additionally, we add as controls the items from our section *General Climate Attitudes & Beliefs* as well as questions included in the "Umweltmodul" in the 2023 SOEP survey[7] and further standard controls (i.e., vote choice, age, gender, east/west, education). This set of control variables will be added to all our analyses and as such we won't explicitly list it further down this section. As the interpretation of logistic regression models is not as straightforward as for the case of linear regression, we will simulate selected quantities of interest and uncertainty (see the approach by King et al., 2000) for all

[7] These include the question asking the respondents worry about the consequences of climate change, people's engagement in climate protection, and general attitudes towards political measures to protect the climate.

our subsequently mentioned analyses. This is not only indispensable in order to publish in international high impact journals, but also to present results in a way which is comprehensible for a larger public audience (see our section *Expected Outreach*).

In a second test for hypotheses 1 to 1c, we employ separate ordered logit models using support for the two undemocratic climate policy items as dependent variables including the same set of predictors and controls as above. For all ordered logit models, we assess the validity of the *proportional odds assumption* using Brant-tests (Brant, 1990). In case of assumption violations, we will estimate partial proportional odds models [PPOMs] that are fit to relax the POA for singular variables only (Williams, 2006).

In order to test hypothesis 2 we use the list experiment in two steps. First, we have to estimate the *hidden support* for the sensitive items in the experiment itself. A simplistic approach would be a comparison of means between treatment and control groups, but we opt to follow the procedure laid out by Graeme and Imai (2012) using their adjacently developed R-package. Their developed estimator enables us to perform multivariate regression to estimate hidden support amongst the whole study population using relevant controls, but also to estimate hidden support for relevant and interesting sub-groups (for which we do not have an expectation, hence this part would be exploratory). In a second step, we then compare the level of hidden agreement predicted within theses models to the level of *open support* for the same sensitive items in the binary agreement battery using difference-in-means estimators (see Robinson and Tannenberg, 2019, Fig. 2 for an example).

In order to test the hypotheses 3, 3a, and 3b, we will again use binary logistic models. In each model, the binary *Economy - Trade-Off* is the dependent variable. The key independent variable for testing hypothesis 3a is the categorical *Economy - Long-Term* item and for testing hypothesis 3b it is the *Economy - Short-Term* item. The control variables are selected as described above. However, it is important to note that we will include a measure of economic security. This measure will be derived from the questions already included in the SOEP regarding household income (i.e., the decile of the income distribution the respondent belongs to). In addition, we will include the various measures of psychological distance as control variables (see *5.Psychological Distance & Emotions*) as we expect that the closer people perceive harmful climate change effects psychologically, the higher the probability that the value economic growth more than climate change mitigation measures. These psychological distance variables will also be included as controls for the testing the remaining two hypotheses.

We will test hypothesis 4 by employing ordered logit models. The dependent variables are constructed based on the specific policy support question in *3. Climate Politics & Policy Preferences*. Regarding the people's personal economic situation, we will again make use of the questions already included in the SOEP regarding household income. In addition, we will control for the same items as described for hypotheses 1, 1a, 1b, and 1c.

Finally, regarding hypothesis 5, we will again estimate ordered logit models with climate policy support as dependent variable. However, this time, we are especially interested in the interaction between climate policies' perceived effectiveness and respondents' expectations about financial consequences of these policies. Again, we include our standard set of control variables.

6.2 Exploratory analysis

We definitely aim to conduct additional exploratory analyses. Most notably, we do not propose a directional hypothesis for the emotions *guilt* and *annoyance*. We will analyse those emotions in the same ways as described above, but without an a priori expected effect. Further, we will explore how all those emotions and different psychological distance items relate to each other (most likely in the form of a factor analysis). If we can extract meaningful factors, we then also re-run our previous analyses with those grouped factor-variables instead of single emotions.

Further, we will examine different predictors for policy support. Specifically, we will look at what factors drive support for different types of climate policies. These types include push- (e.g., taxes) and pull-measures (e.g., subsidies) as well as regulations (e.g., duties or limits). This would require the estimation of ordered logit models. Subsequently, we will measure what factors predict the support for all of these policy types compared to only support for specific types (i.e., what drives people to support the full spectrum of climate policies?). Based on our proposed questions on specific climate policies, we are able to construct different policy profiles (e.g., support for all policy types, only support for push-measures, no support for any types, etc.). Predictors of such profiles can be estimated by the use of multinomial choice models. At this stage, we have no expectations about effect size or direction.

7 Expected Outreach

We aim to publish the analyses based on this module as research articles in international peer-reviewed, high impact journals (e.g., Journal of Politics, Environment and Behavior, Environmental Politics, British Journal of Political Science). We aim at a minimum of two articles targeting our direct research questions, one focusing on the economic, one on the democratic trade-off argument.

Beyond that, we think the module has a lot of potential for secondary analyses by other researchers (but also by us eventually), especially in combination with the already planned "Umweltmodel" in the 2023 SOEP survey. This could include, but is not limited to the following research questions:

- How do emotions and psychological distance towards climate change affect climate-related behavior (e.g., travel, living and eating practices)?
- Does valuing environmental over economic goals on a societal policy level directly lead to pro-climate behavior and potentially costly investments into energy-saving appliances on the individual level?
- How do emotions and psychological distance towards climate change affect whether you participate in the public discourse on climate change?
- Creating climate-attitude based typologies of the German public modeled after the *Six Americas* (Roser-Renouf et al., 2009).
- Why and when does abstract support for climate policy (types) translate into support for concrete policies?
- How do climate attitudes and the value conflict between economic, democratic and environmental goals translate into voting behavior?
- Additionally, the SOEP data could also allow for any kind of spatial analyses: In which areas are citizens more or less concerned about climate change? Does this (or regional peculiarities in policy support) affect regional election or policy outcomes? Can they be explained by regional level factors?

Furthermore, both the raw data and the results of our analyses should be of great interest to the international and national press. For example, The Economist (2022) recently reported on the results of a survey regarding the scientific consensus on climate change. The SPIEGEL (2022) reported on a British survey on public perceptions on climate change. We strongly believe that our module also has the potential to get media coverage in such high profile outlets, especially since our selected policies are rooted in current debates (e.g., subsidies for public transport, mandatory solar panels). This, in turn, would increase the visibility of the SOEP-IS and the SOEP in general, respectively.

8 Estimation of Total Interview Time

We programmed our proposed module using *Qualtrics*, where the algorithm estimated a total interview time of **7 minutes and 42 seconds**.

9 Content of Questions

9.1 German Version

Democratic Trade-Offs: List Experiment

1. **List Experiment - Sensitive Group I** Hier sind einige Aussagen denen man zustimmen kann oder nicht. Bitte sagen Sie uns **wie vielen** dieser Aussagen Sie zustimmen würden. Bitte sagen Sie uns nicht welchen genau Sie zustimmen, sondern nur wie vielen davon.

   (a) In Zeiten einer Pandemie wie Corona sollte es eine allgemeine Impfpflicht geben.

   (b) Die Regierung sollte Maßnahmen ergreifen, um die Einkommensunterschiede zu verringern.

   (c) Die europäische Einigung sollte nicht weiter vorangetrieben werden.

   (d) Noch stärkere Klimabekämpfungs-Maßnahmen sollten notfalls auch mit undemokratischen Mitteln verhindert werden.

   - Scale:
     a. 0
     b. 1
     c. 2
     d. 3
     e. 4

2. **List Experiment - Sensitive Group II** Hier sind einige Aussagen denen man zustimmen kann oder nicht. Bitte sagen Sie uns **wie vielen** dieser Aussagen Sie zustimmen würden. Bitte sagen Sie uns nicht welchen genau Sie zustimmen, sondern nur wie vielen davon.

   (a) In Notlagen ist es gerechtfertigt, dass der Staat Freiheitsrechte der Bürgerinnen und Bürger einschränkt.

   (b) Die Regierung sollte einen landesweiten Mietendeckel einführen.

   (c) Der Staat sollte sich aus der Wirtschaft heraushalten.

   (d) Stärkere Klimabekämpfungs-Maßnahmen sollten notfalls auch mit undemokratischen Mitteln durchgesetzt werden.

   - Scale:
     a. 0
     b. 1
     c. 2
     d. 3
     e. 4

3. **List Experiment - Control Group I** Hier sind einige Aussagen denen man zustimmen kann oder nicht. Bitte sagen Sie uns **wie vielen** dieser Aussagen Sie zustimmen würden. Bitte sagen Sie uns nicht welchen genau Sie zustimmen, sondern nur wie vielen davon.

   (a) In Notlagen ist es gerechtfertigt, dass der Staat Freiheitsrechte der Bürgerinnen und Bürger einschränkt.

   (b) Die Regierung sollte einen landesweiten Mietendeckel einführen.

   (c) Der Staat sollte sich aus der Wirtschaft heraushalten.

   - Scale:
     a. 0
     b. 1
     c. 2
     d. 3

4. **List Experiment - Control Group II** Hier sind einige Aussagen denen man zustimmen kann oder nicht. Bitte sagen Sie uns **wie vielen** dieser Aussagen Sie zustimmen würden. Bitte sagen Sie uns nicht welchen genau Sie zustimmen, sondern nur wie vielen davon.

   (a) In Zeiten einer Pandemie wie Corona sollte es eine allgemeine Impfpflicht geben.

   (b) Die Regierung sollte Maßnahmen ergreifen, um die Einkommensunterschiede zu verringern.

   (c) Die europäische Einigung sollte nicht weiter vorangetrieben werden.

   - Scale:
     a. 0
     b. 1
     c. 2
     d. 3

General Climate Attitudes & Beliefs

5. **Existence** Es wird viel darüber gesprochen, ob sich das Klima weltweit ändert. Wie ist Ihre Meinung dazu? Glauben Sie, dass es den Klimawandel gibt?

   (a) Sehr sicher, dass es den Klimawandel gibt

   (b) Eher sicher, dass es den Klimawandel gibt

   (c) Bin mir unsicher

   (d) Eher sicher, dass es den Klimawandel nicht gibt

   (e) Sehr sicher, dass es den Klimawandel nicht gibt

6. **Cause** Glauben Sie, dass der Klimawandel eine Folge natürlicher Prozesse oder eine Folge menschlichen Handelns ist?

(a) 1 (ausschließlich Folge natürlicher Prozesse)
(b) 2
(c) 3
(d) 4
(e) 5
(f) 6 (zu gleichen Teilen Folge natürlicher Prozesse und menschlichen Handelns)
(g) 7
(h) 8
(i) 9
(j) 10
(k) 11 (ausschließlich Folge menschlichen Handelns)
(l) Weiß nicht
(m) Es gibt keinen Klimawandel

7. **Information** Wie gut fühlen Sie sich selbst über das Thema Klimawandel informiert?

(a) Sehr gut informiert
(b) Eher gut informiert
(c) Teils / teils
(d) Eher schlecht informiert
(e) Sehr schlecht informiert

Climate Politics & Policy Preferences

8. **Climate Politics - Attitudes** Wie ist Ihre allgemeine Meinung zur Bekämpfung des Klimawandels?

(a) 1 (Die Politik muss viel mehr zur Bekämpfung des Klimawandels tun)
(b) 2
(c) 3
(d) 4
(e) 5
(f) 6 (Die Politik zur Bekämpfung des Klimawandels ist genau richtig)
(g) 7
(h) 8
(i) 9
(j) 10
(k) 11 (Die Politik zur Bekämpfung des Klimawandels geht schon viel zu weit)
(l) Weiß nicht

Wir würden Sie gerne zu konkreten politischen Maßnahmen befragen. Inwieweit stimmen Sie zu, dass die folgenden Maßnahmen zur Bekämpfung des Klimawandels ergriffen werden sollten?

9. **Policy - Pull I** Höhere Subventionierung des öffentlichen Nahverkehrs (Bus, Straßenbahn, U-Bahn) aus allgemeinen Steuermitteln.

   (a) Stimme komplett zu
   (b) Stimme eher zu
   (c) Teils/teils
   (d) Stimme eher nicht zu
   (e) Stimme komplett nicht zu
   (f) Weiß nicht

10. **Policy - Pull II** Höhere Subventionierung des Kaufs von Elektro-Autos aus allgemeinen Steuermitteln.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

11. **Policy - Push I** Einführung einer Unternehmenssteuer, deren Höhe sich nach dem CO2-Ausstoß richtet.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

12. **Policy - Push II** Höhere Besteuerung von fossilen Brennstoffen wie Benzin oder Heizöl.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

13. **Policy - Regulation I** Einführung eines allgemeinen Tempolimits von 130 auf Autobahnen.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

14. **Policy - Regulation II** Einführung einer Solarpflicht (Installation von Solaranlagen) auf allen Neubauten (Gewerbe und private Wohnhäuser).

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

15. **Policy - Democratic I** Entfernen aller Beiträge und Kommentare im Internet und sozialen Medien, in denen gesagt wird, es gäbe keinen Klimawandel.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

16. **Policy - Democratic II** Einräumen neuer Kompetenzen für die Regierung, Grundbesitz (privat und gewerblich) zu verstaatlichen, um den Ausbau erneuerbarer Energien (z.B. Wind- oder Solarparks) voranzutreiben.

    (a) Stimme komplett zu
    (b) Stimme eher zu
    (c) Teils/teils
    (d) Stimme eher nicht zu
    (e) Stimme komplett nicht zu
    (f) Weiß nicht

17. **Benefits** Wählen Sie alle Maßnahmen aus, welche Ihrer Meinung nach finanzielle Vorteile für Sie persönlich mit sich bringen würden.

(a) Subventionierung des öffentlichen Nahverkehr

(b) Subventionierung des Kaufs von Elektro-Autos

(c) Neue CO2-Steuern für Unternehmen

(d) Höhere Besteuerung von fossilen Brennstoffen

(e) Einführung Tempolimit 130

(f) Einführung einer Solarpflicht

(g) Entfernen aller Beiträge, die sagen, es gibt keinen Klimawandel

(h) Neue Regierungs-Kompetenzen, Grundbesitz für Ausbau erneuerbarer Energien zu verstaatlichen

18. **Costs** Wählen Sie alle Maßnahmen aus, welche Ihrer Meinung nach finanzielle Nachteile für Sie persönlich mit sich bringen würden.

(a) Subventionierung des öffentlichen Nahverkehr

(b) Subventionierung des Kaufs von Elektro-Autos

(c) Neue CO2-Steuern für Unternehmen

(d) Höhere Besteuerung von fossilen Brennstoffen

(e) Einführung Tempolimit 130

(f) Einführung einer Solarpflicht

(g) Entfernen aller Beiträge, die sagen, es gibt keinen Klimawandel

(h) Neue Regierungs-Kompetenzen, Grundbesitz für Ausbau erneuerbarer Energien zu verstaatlichen

19. **Effectiveness** Wählen Sie alle Maßnahmen aus, welche Ihrer Meinung nach effektiv zur Bekämpfung des Klimawandels beitragen würden.

(a) Subventionierung des öffentlichen Nahverkehr

(b) Subventionierung des Kaufs von Elektro-Autos

(c) Neue CO2-Steuern für Unternehmen

(d) Höhere Besteuerung von fossilen Brennstoffen

(e) Einführung Tempolimit 130

(f) Einführung einer Solarpflicht

(g) Entfernen aller Beiträge, die sagen, es gibt keinen Klimawandel

(h) Neue Regierungs-Kompetenzen, Grundbesitz für Ausbau erneuerbarer Energien zu verstaatlichen

Economic Trade-Offs

Es gibt unterschiedliche Auffassungen über die kurzfristigen und langfristigen wirtschaftlichen Folgen von Maßnahmen zur Bekämpfung des Klimawandels.

20. **Economy - Short-Term** . Was glauben Sie, wie wirken sich Maßnahmen zur Bekämpfung des Klimawandels kurzfristig auf die deutsche Wirtschaft und den Arbeitsmarkt aus?

(a) Sehr positiv
(b) Eher positiv
(c) Gleichermaßen positiv wie negativ
(d) Eher negativ
(e) Sehr negativ
(f) Gar nicht

21. **Economy - Long-Term** . Was glauben Sie, wie wirken sich Maßnahmen zur Bekämpfung des Klimawandels langfristig auf die deutsche Wirtschaft und den Arbeitsmarkt aus?

(a) Sehr positiv
(b) Eher positiv
(c) Gleichermaßen
(d) Eher negativ
(e) Sehr negativ
(f) Gar nicht

22. **Economy - Trade-Off** Stellen Sie sich vor, dass es bei einer konkreten Maßnahme zu einem Konflikt zwischen der Bekämpfung des Klimawandels und dem Wirtschaftswachstum kommt. Was ist Ihrer Meinung nach nun wichtiger?

(a) Wirtschaftswachstum, auch wenn dies negative Auswirkungen auf das Klima hat.
(b) Bekämpfung des Klimawandels, auch wenn dies negative Auswirkungen auf das Wirtschaftswachstum hat.

Psychological Distance & Emotions

23. **PD - Personal** Wie stark wird sich der Klimawandel Ihrer Meinung nach negativ auf Sie persönlich auswirken? [Note: extra space between first four answers and "Gar nicht".]

(a) Sehr stark
(b) Eher stark
(c) Eher schwach
(d) Sehr schwach
(e) Gar nicht

24. **PD - Local** Wie stark wird sich der Klimawandel Ihrer Meinung nach negativ auf Ihre Gegend auswirken? [Note: extra space between first four answers and "Gar nicht".]

(a) Sehr stark
(b) Eher stark
(c) Eher schwach
(d) Sehr schwach
(e) Gar nicht

25. **PD - Country** Wie stark wird sich der Klimawandel Ihrer Meinung nach negativ auf Deutschland auswirken? [Note: extra space between first four answers and "Gar nicht".]

(a) Sehr stark
(b) Eher stark
(c) Eher schwach
(d) Sehr schwach
(e) Gar nicht

26. **PD - World** Wie stark wird sich der Klimawandel Ihrer Meinung nach negativ auf Länder des globalen Südens auswirken? [Note: extra space between first four answers and "Gar nicht".]

(a) Sehr stark
(b) Eher stark
(c) Eher schwach
(d) Sehr schwach
(e) Gar nicht

27. **PD - Timing Germany** Wann denken Sie werden negative Folgen des Klimawandels in Deutschland zu spüren sein?

(a) Negativen Folgen sind bereits spürbar
(b) Negativen Folgen werden in [OPEN ANSWER - NUMBER OF YEARS] spürbar sein.
(c) Negativen Folgen werden nie spürbar sein

28. **PD - Timing World** Wann denken Sie werden negative Folgen des Klimawandels weltweit zu spüren sein?

(a) Negative Folgen sind bereits spürbar
(b) Negative Folgen werden in [OPEN ANSWER - NUMBER OF YEARS] spürbar sein.
(c) Negative Folgen werden nie spürbar sein

29. **Emotions - Climate Change** Wie stark empfinden Sie die folgenden Emotionen, wenn Sie an den Klimawandel und seine möglichen Folgen denken. [Note: extra space between first four answers and "Gar nicht".]

(a) Wut / wütend

(b) Angst / ängstlich

(c) Gelassenheit / gelassen

(d) Schuld / schuldig

(e) Genervtheit / genervt

(f) Zuversicht / zuversichtlich

(g) Hoffnungslosigkeit / hoffnungslos

(h) i. Scale:
   a. Sehr Stark
   b. Eher Stark
   c. Eher Schwach
   d. Sehr Schwach
   e. Gar nicht

30. **Emotions - Government** Und wie stark empfinden Sie die folgenden Emotionen, wenn Sie explizit an die Politik denken, die derzeit in Sachen Klimawandel gemacht wird. [Note: extra space between first four answers and "Gar nicht".]

(a) Wut / wütend

(b) Angst / ängstlich

(c) Gelassenheit / gelassen

(d) Schuld / schuldig

(e) Genervtheit / genervt

(f) Zuversicht / zuversichtlich

(g) Hoffnungslosigkeit / hoffnungslos

(h) i. Scale:
   a. Sehr Stark
   b. Eher Stark
   c. Eher Schwach
   d. Sehr Schwach
   e. Gar nicht

Democratic Trade-Offs: Direct

31. **Undemocratic - Direct** [Partially based on List Experiment Randomization] Hier sind einige Aussagen denen man zustimmen kann oder nicht. Bitte wählen Sie jede dieser Aussagen aus, der Sie persönlich zustimmen würden.

(a) Noch stärkere Klimabekämpfungs-Maßnahmen sollten notfalls auch mit undemokratischen Mitteln verhindert werden. [Only Ex-Group: I]

(b) Stärkere Klimabekämpfungs-Maßnahmen sollten notfalls auch mit undemokratischen Mitteln durchgesetzt werden. [Only Ex-Group: II]

(c) Die Klimapolitik sollte von Wissenschaftlern und Experten anstelle von gewählten Politiker entschieden werden.

(d) Politiker sollten nur ein Mandat im Bundestag erhalten dürfen, wenn Sie öffentlich anerkennen, dass der Klimwandel existiert und menschengemacht ist.

(e) Proteste von Klimaaktivisten, die das alltägliche Leben beeinträchtigen, sollten verboten und die Aktivisten angezeigt werden.

(f) Selbst wenn Gerichte urteilen, dass stärkere Klimabekämpfungsmaßnahmen ergriffen werden müssen, sollte die Politik sich darüber hinwegsetzen.

- Scale:
  a. Ja
  b. Nein

9.2 English Version

Democratic Trade-Offs: List Experiment

1. **List Experiment - Sensitive Group I** Here are some statements you may or may not agree with. Please tell us **how many** of these statements you would agree with. Please do not tell us which ones exactly you agree with, just how many of them.

(a) In times of a pandemic like Corona, there should be a universal vaccination requirement.

(b) The government should take action to reduce income inequality.

(c) European integration should not be advanced any further.

(d) If necessary, stronger climate control measures should be prevented by undemocratic means

- Scale:
  a. 0
  b. 1
  c. 2
  d. 3
  e. 4

2. **List Experiment - Sensitive Group II** Here are some statements you may or may not agree with. Please tell us **how many** of these statements you would agree with. Please do not tell us which ones exactly you agree with, just how many of them.

(a) In emergency situations, it is justified for the state to restrict citizens' civil liberties.

(b) The government should implement a nationwide rent cap.

(c) The state should refrain from getting involved in the economy.

(d) If necessary, stronger climate control measures should be introduced by undemocratic means

- Scale:
  a. 0
  b. 1
  c. 2
  d. 3
  e. 4

3. **List Experiment - Control Group I** Here are some statements you may or may not agree with. Please tell us **how many** of these statements you would agree with. Please do not tell us which ones exactly you agree with, just how many of them.

(a) In emergency situations, it is justified for the state to restrict citizens' civil liberties.

(b) The government should implement a nationwide rent cap.

(c) The state should refrain from getting involved in the economy.

- Scale:
  a. 0
  b. 1
  c. 2
  d. 3

4. **List Experiment - Control Group II** Here are some statements you may or may not agree with. Please tell us **how many** of these statements you would agree with. Please do not tell us which ones exactly you agree with, just how many of them.

(a) In times of a pandemic like Corona, there should be a universal vaccination requirement.

(b) The government should take action to reduce income inequality.

(c) European integration should not be advanced any further.

- Scale:
  a. 0
  b. 1
  c. 2
  d. 3

General Climate Attitudes & Beliefs

5. **Existence** There is a lot of talk about whether the climate is changing around the world. What is your opinion on this? Do you believe that climate change is happening?

(a) Very sure that climate change exists

(b) Rather sure that climate change exists

(c) I am unsure

(d) Rather sure that climate change does not exist

(e) Very sure that climate change does not exist

6. **Cause** Do you think climate change is a result of natural processes or a result of human activity?

(a) 1 (exclusively result of natural processes)

(b) 2

(c) 3

(d) 4

(e) 5

(f) 6 (in equal parts the result of natural processes and human activity)

(g) 7

(h) 8

(i) 9

(j) 10

(k) 11 (exclusively the result of human action)

(l) Don't know

(m) There is no climate chnange

7. **Information** How well informed do you feel yourself to be on the topic of climate change?

(a) Very well informed

(b) Rather well informed

(c) Partly well / partly poorly informed

(d) Rather poorly informed

(e) Very poorly informed

Climate Politics & Policy Preferences

8. **Climate Politics - Attitudes** What is your general opinion on combating climate change?

(a) 1 (Policies to combat climate change are not enough)
(b) 2
(c) 3
(d) 4
(e) 5
(f) 6 (Policies to combat climate change are spot on)
(g) 7
(h) 8
(i) 9
(j) 10
(k) 11 (Policies to combat climate change already go far too far)
(l) Don't know

We would like to ask you about specific policy measures. To what extent do you agree that the following measures should be taken to combat climate change?

9. **Policy - Pull I** Higher subsidies for public transport (bus, tram, metro) from general tax revenues.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

10. **Policy - Pull II** Higher subsidies for the purchase of electric cars from general tax revenues.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

11. **Policy - Push I** Introduction of a corporate tax, the amount of which is based on CO2 emissions.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

12. **Policy - Push II** Higher taxation of fossil fuels such as gasoline or heating oil.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

13. **Policy - Regulation I** Introduction of a general speed limit of 130 on highways.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

14. **Policy - Regulation II** Introduce a solar duty (installation of solar panels) for all new buildings (commercial and private residences).

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

15. **Policy - Democratic I** Remove all posts and comments on the Internet and social media that say there is no climate change.

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

16. **Policy - Democratic II** Granting new powers to the government to nationalize land holdings (private and commercial) to drive renewable energy development (e.g., wind or solar farms).

(a) Completely agree
(b) Rather agree
(c) Partly agree / partly disagree
(d) Rather disagree
(e) Completely disagree
(f) Don't know

17. **Benefits** Select all measures which you think would bring financial benefits for you personally.

(a) Subsidizing public transport
(b) Subsidizing the purchase of electric cars
(c) New CO2 taxes for companies
(d) Higher taxation of fossil fuels
(e) Introduction of speed limit of 130
(f) Introduction of a solar duty
(g) Remove all posts that say there is no climate change
(h) New government competencies to nationalize land ownership for renewable energy development

18. **Costs** Select all measures, which in your opinion would bring financial disadvantages for you personally.

(a) Subsidizing public transport
(b) Subsidizing the purchase of electric cars
(c) New CO2 taxes for companies
(d) Higher taxation of fossil fuels
(e) Introduction of speed limit of 130
(f) Introduction of a solar duty
(g) Remove all posts that say there is no climate change
(h) New government competencies to nationalize land ownership for renewable energy development

19. **Effectiveness** Select all the measures that you think would be effective in combating climate change.

- (a) Subsidizing public transport
- (b) Subsidizing the purchase of electric cars
- (c) New CO2 taxes for companies
- (d) Higher taxation of fossil fuels
- (e) Introduction of speed limit of 130
- (f) Introduction of a solar duty
- (g) Remove all posts that say there is no climate change
- (h) New government competencies to nationalize land ownership for renewable energy development

Economic Trade-Offs

There are different views on the short-term and long-term economic consequences of climate change mitigation measures.

20. **Economy - Short-Term** . How do you think measures to combat climate change will affect the German economy and labor market in the short term?

- (a) Very positive
- (b) Rather positive
- (c) Equally positive and negative
- (d) Rather negative
- (e) Very negative
- (f) Not at all

21. **Economy - Long-Term** . How do you think measures to combat climate change will affect the German economy and labor market in the long term?

- (a) Very positive
- (b) Rather positive
- (c) Equally positive and negative
- (d) Rather negative
- (e) Very negative
- (f) Not at all

22. **Economy - Trade-Off** Imagine that a specific measure involves a conflict between combating climate change and economic growth. Which do you think is more important?

- (a) Economic growth, even if this has a negative impact on the climate.
- (b) Combating climate change, even if this has a negative impact on economic growth.

Psychological Distance & Emotions

23. **PD - Personal** How much do you think climate change will negatively impact you personally? [Note: extra space between first four answers and "Not at all".]

    (a) Very strong
    (b) Rather strong
    (c) Rather weak
    (d) Very weak
    (e) Not at all

24. **PD - Local** How much do you think climate change will negatively impact your area? [Note: extra space between first four answers and "Not at all".]

    (a) Very strong
    (b) Rather strong
    (c) Rather weak
    (d) Very weak
    (e) Not at all

25. **PD - Country** How much of a negative impact do you think climate change will have on Germany? [Note: extra space between first four answers and "Not at all".]

    (a) Very strong
    (b) Rather strong
    (c) Rather weak
    (d) Very weak
    (e) Not at all

26. **PD - World** How much of a negative impact do you think climate change will have on countries in the global South? [Note: extra space between first four answers and "Not at all".]

    (a) Very strong
    (b) Rather strong
    (c) Rather weak
    (d) Very weak
    (e) Not at all

27. **PD - Timing Germany** When do you think negative consequences of climate change will be felt in Germany?

(a) Negative consequences are already felt

(b) Negative consequences will be felt in [OPEN ANSWER - NUMBER OF YEARS] years

(c) Negative consequences will never be felt

28. **PD - Timing World** When do you think negative consequences of climate change will be felt worldwide?

(a) Negative consequences are already felt

(b) Negative consequences will be felt in [OPEN ANSWER - NUMBER OF YEARS] years

(c) Negative consequences will never be felt

29. **Emotions - Climate Change** How strongly do you feel the following emotions when you think about climate change and its possible consequences?

(a) Anger / angry

(b) Fear / fearful

(c) Serenity / serene

(d) Guilt / guilty

(e) Annoyed / annoyed

(f) Optimism / optimistic

(g) Hopelessness / hopeless

(h) i. Scale:
   a. Very strong
   b. Rather strong
   c. Rather weak
   d. Very weak
   e. Not at all

30. **Emotions - Government** And how strongly do you feel the following emotions when you think explicitly about the policies that are currently being made about climate change?

(a) Anger / angry

(b) Fear / fearful

(c) Serenity / serene

(d) Guilt / guilty

(e) Annoyed / annoyed

(f) Optimism / optimistic

(g) Hopelessness / hopeless

(h) i. Scale:
   a. Very strong
   b. Rather strong
   c. Rather weak
   d. Very weak
   e. Not at all

Democratic Trade-Offs: Direct

31. **Undemocratic - Direct** [Partially based on List Experiment Randomization] Here are some statements you may or may not agree with. Please select each of these statements that you would personally agree with.

(a) Stronger measures to combat climate change should be prevented by undemocratic means if necessary. [Only Ex-Group: I]

(b) Stronger measures to combat climate change should be implemented by undemocratic means if necessary. [Only Ex-Group: II]

(c) Climate politics should be decided by scientists and experts instead of elected politicians.

(d) Politicians should only be allowed to hold a mandate in the Bundestag if they publicly acknowledge that climate change exists and is man-made.

(e) Protests by climate activists that interfere with everyday life should be banned and the activists reported.

(f) Even if courts rule that stronger climate action must be taken, politicians should override this.

- Scale:
  a. Yes
  b. No

References

Aklin, M. and J. Urpelainen (2013). Debating clean energy: Frames, counter frames, and audiences. *Global Environmental Change 23*(5), 1225–1232.

Amat, F., A. Arenas, A. Falcó-Gimeno, and J. Muñoz (2020). Pandemics meet democracy. experimental evidence from the covid-19 crisis in spain.

Ansolabehere, S. and D. M. Konisky (2014). *Cheap and clean: how Americans think about energy in the age of global warming*. Mit Press.

Armingeon, K. and R. Bürgisser (2021). Trade-offs between redistribution and environmental protection: the role of information, ideology, and self-interest. *Journal of European Public Policy 28*(4), 489–509.

Brant, R. (1990). Assessing proportionality in the proportional odds model for ordinal logistic regression. *Biometrics*, 1171–178.

Brulle, R. J., J. Carmichael, and J. C. Jenkins (2012). Shifting public opinion on climate change: an empirical assessment of factors influencing concern over climate change in the us, 2002–2010. *Climatic change 114*(2), 169–188.

Chu, H. and J. Z. Yang (2019). Emotion and the psychological distance of climate change. *Science Communication 41*(6), 761–789.

Cologna, V. and M. Siegrist (2020). The role of trust for climate change mitigation and adaptation behaviour: A meta-analysis. *Journal of Environmental Psychology 69*, 101428.

Conrad, K. (2019). Climate change preferences and values. *SSRN*.

Ding, D., E. W. Maibach, X. Zhao, C. Roser-Renouf, and A. Leiserowitz (2011). Support for climate policy and societal action are linked to perceptions about scientific agreement. *Nature Climate Change 1*(9), 462–466.

DPA/ZDF (2022). Czaja will Blockierer präventiv einsperren.

DPI (2022). Democracy perception index. Technical report.

Drews, S. and J. C. van den Bergh (2016). What explains public support for climate policies? a review of empirical and experimental studies. *Climate Policy 16*(7), 855–876.

Dunwoody, P. T. and S. G. McFarland (2018). Support for anti-muslim policies: The role of political traits and threat perception. *Political Psychology 39*(1), 89–106.

Economist (2022). Which european countries know the least about climate change?

EIB (2021). The climate crisis in a covid-19 world: Calls for a green recovery. Technical report.

ESS (2016). Ess round 8: European social survey round 8 data. data file edition 2.1. nsd-norwegian centre for research data, norway–data archive and distributor of ess data for ess eric. doi: 10.21338. Technical report, NSD-ESS8-2016.

Gellrich, A. (2020). Umweltbewusstsein in deutschland 2020. Technical report.

GLES (2022). Gles cross-section 2021, pre-election. GESIS Data Archive, Cologne. Data file version 2.0.0, doi:10.4232/1.13860.

Goldberg, M., S. V. D. Linden, A. Leiserowitz, and E. Maibach (2019). Perceived social consensus can reduce ideological biases on climate change. *Environment and Behaviour 52*(5), 495–517.

Graeme, B. and K. Imai (2012). Statistical analysis of list experiments. *Political Analysis 20*, 47–77.

Graham, M. H. and M. W. Svolik (2020). Democracy in america? partisanship, polarization, and the robustness of support for democracy in the united states. *American Political Science Review 114*(2), 392–409.

Hammar, H. and S. C. Jagers (2007). What is a fair co2 tax increase? on fair emission reductions in the transport sector. *Ecological Economics 61*(2), 377–387.

Huber, R. A. and M. Wicki (2021). What explains citizen support for transport policy? the roles of policy design, trust in government and proximity among swiss citizens. *Energy Research & Social Science 75*, 101973.

Huber, R. A., M. L. Wicki, and T. Bernauer (2020). Public support for environmental policy depends on beliefs concerning effectiveness, intrusiveness, and fairness. *Environmental Politics 29*(4), 649–673.

Jagers, S. C. and H. Hammar (2009). Environmental taxation for good and for bad: the efficiency and legitimacy of sweden's carbon tax. *Environmental Politics 18*(2), 218–237.

Kachi, A., T. Bernauer, and R. Gampfer (2015). Climate policy in hard times: are the pessimists right? *Ecological Economics 114*, 227–241.

Kahn, M. E. and M. J. Kotchen (2011). Business cycle effects on concern about climate change: the chilling effect of recession. *Climate Change Economics 2*(03), 257–273.

Kellstedt, P. M., M. D. Ramirez, A. Vedlitz, and S. Zahran (2019). Does political sophistication minimize value conflict? evidence from a heteroskedastic graded irt model of opinions toward climate change. *British Journal of Political Science 49*(4), 1309–1332.

Kiewiet de Jonge, C. P. (2016, 05). Should Researchers Abandon Questions about "Democracy"?: Evidence from Latin America. *Public Opinion Quarterly 80*(3), 694–716.

King, G., M. Tomz, and J. Wittenberg (2000). Making the most of statistical analyses: Improving interpretation and presentation. *American journal of political science*, 347–361.

Kulin, J., I. J. Sevä, and R. E. Dunlap (2021, March). Nationalist ideology, rightwing populism, and public views about climate change in europe. *Environmental Politics 30*(7), 1111–1134.

Liberman, N. and Y. Trope (2008). The psychology of transcending the here and now. *Science 322*(5905), 1201–1205.

Maiella, R., P. L. Malva, D. Marchetti, E. Pomarico, A. D. Crosta, R. Palumbo, L. Cetara, A. D. Domenico, and M. C. Verrocchio (2020, November). The psychological distance and climate change: A systematic review on the mitigation and adaptation behaviors. *Frontiers in Psychology 11*.

Mayer, A. and E. K. Smith (2019). Unstoppable climate change? the influence of fatalistic beliefs about climate change on behavioural change and willingness to pay cross-nationally. *Climate Policy 19*(4), 511–523.

Mildenberger, M. and A. Leiserowitz (2017). Public opinion on climate change: Is there an economy–environment tradeoff? *Environmental Politics 26*(5), 801–824.

Mittiga, R. (2021). Political legitimacy, authoritarianism, and climate change. *American Political Science Review*, 1–14.

Peritia (2022). Public perceptions on climate change. Technical report.

Robinson, D. and M. Tannenberg (2019). Self-censorship of regime support in authoritarian states: Evidence from list experiments in china. *Research & Politics 6*(3), 2053168019856449.

Roser-Renouf, C., E. Maibach, and A. Leiserowitz (2009, 12). Global warming's six americas: An audience segmentation analysis (invited). *AGU Fall Meeting Abstracts*.

Schneider, C. R., L. Zaval, and E. M. Markowitz (2021). Positive emotions and climate change. *Current Opinion in Behavioral Sciences 42*, 114–120. Human Response to Climate Change: From Neurons to Collective Action.

Shepherd, L., F. Fasoli, A. Pereira, and N. R. Branscombe (2018). The role of threat, emotions, and prejudice in promoting collective action against immigrant groups. *European Journal of Social Psychology 48*(4), 447–459.

Singh, A. S., A. Zwickle, J. T. Bruskotter, and R. Wilson (2017). The perceived psychological distance of climate change impacts and its influence on support for adaptation policy. *Environmental Science & Policy 73*, 93–99.

Smith, N. and A. Leiserowitz (2014). The role of emotion in global warming policy support and opposition. *Risk Analysis 34*(5), 937–948.

Spence, A., W. Poortinga, and N. Pidgeon (2012). The psychological distance of climate change. *Risk Analysis 32*(6), 957–972.

Stöcker, C. (2022). Umgang mit der klimakrise - was gegen die verdrängung hilft. *Spiegel Wissenschaft*.

Trope, Y. and N. Liberman (2010). Construal-level theory of psychological distance. *Psychological Review 117*, 440–463.

Valentino, N. A., T. Brader, E. W. Groenendyk, K. Gregorowicz, and V. L. Hutchings (2011, January). Election night's alright for fighting: The role of emotions in political participation. *The Journal of Politics 73*(1), 156–170.

Valkengoed, A. M. V. and L. Steg (2019). Meta-analyses of factors motivating climate change adaptation behaviour. *Nature Climate Change 9*(2), 158–163.

van Doorn, J. (2018). Anger, feelings of revenge, and hate. *Emotion Review 10*(4), 321–322.

Wagner, M. and D. Morisi (2019, November). Anxiety, fear, and political decision making.

Wang, S., M. J. Hurlstone, Z. Leviston, I. Walker, and C. Lawrence (2019, February). Climate change from a distance: An analysis of construal level and psychological distance from climate change. *Frontiers in Psychology 10*.

Wang, S., Z. Leviston, M. Hurlstone, C. Lawrence, and I. Walker (2018). Emotions predict policy support: Why it matters how people feel about climate change. *Global Environmental Change 50*, 25–40.

Wicki, M. and L. Fesenfeld (2019). In search of politically feasible policy-packages for sustainable passenger transport: insights from choice experiments in china, germany, and the usa. *Environmental Research Letters 14*.

Williams, R. (2006). Generalized ordered logit/partial proportional odds models for ordinal dependent variables. *The Stata Journal 6*, 58–82.

Zahran, S., S. D. Brody, H. Grover, and A. Vedlitz (2006). Climate change vulnerability and policy support. *Society & Natural Resources 19*(9), 771–789.

Zweck, B. and M. Rathje (2021). Soep-is 2020 – survey report on the 2020 soep innovation sample. Technical report.